# Essays

|!|BCEEB0110000001|!| ★ **Cross topic**

Both humans and plants develop vascular system(s) for transport within their bodies. Contrast the structure of their vascular systems. Explain why transport in human is more effective than that in plants. (11 marks)

##

**Structures of their vascular systems:** (Max. 4)

- Humans have two connected systems—the lymphatic system opens into the blood circulation, while plants have two separate systems—xylem and phloem. (1)
- The blood vessels and lymph vessels in humans are made up of living cells. In plants, xylem is made up of dead cells and phloem is made up of living cells. (1)
- The lumens of vessels in humans are surrounded by cells (extracellular), while the lumens of vessels in plants are within cells (intracellular). (1)
- The diameter of vessels in humans can be adjusted (arteries are elastic / veins can be distended), while vessels in plants are rigid and their diameter is fixed. (1)
- Valves are present in veins and lymph vessels of humans, while sieve plates are present in the phloem of plants. (1)

**Why transport in humans is more effective:** (Max. 4)

- In humans, the heart acts as a pump to provide the driving force to propel blood along vessels, while there is no well defined organ to generate a driving force in plants. (1)
- In humans, the continuous flow of blood and lymph are maintained by various ways:
  - the elastic wall of arteries recoils and pushes blood forward (1)
  - presence of valves in veins and lymph vessels to prevent backflow (1)
  - contraction of skeletal muscle squeezes lymph and venous blood towards the heart (1)
  - inhalation creates a negative pressure in the thorax to draw in lymph and venous blood from the peripheral circulation (1)

  while transpiration pull and root pressure in plants provide the driving force to maintain the flow. (1)

**Communication** (Max. 3)

(11 marks)

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| Skill assessed | Marks |
|---|---|
| Understanding basic concepts | 8 |
| Communication | 3 |

|!|BCEEA0110000002|!| ★★ **Cross topic**

Describe the similarities and differences in the structure and functioning of xylem vessels in vascular plants and arteries in humans. (13 marks)

##

**Similarities:** (Max. 2)

- Both are tubular structures / contain a lumen. (1)
- Both are responsible for the transport of liquid within the organisms. (1)
- The liquids in both structures move in one direction only. (1)

(Or other acceptable answers)

**Differences:** (any ***four*** pairs of the following) (Max. 8)

| Xylem vessels | Arteries |
|---|---|
| Transport water (and minerals) (1) | Transport blood (1) |
| Dead cells (1) | Made up of living cells (1) |
| No pumping organ (1) | Pressure generated by a pumping organ: heart (1) |
| Provide support to the organisms (1) | Do not provide support to the organisms (1) |
| Their walls contain only lignin (1) | Their walls have muscle / elastic tissue (1) |
| The movement of liquid is smooth (1) | The movement of liquid is in pulses (1) |

(Or other acceptable answers)

**Communication** (Max. 3)

(13 marks)

##

| Skill assessed | Marks |
|---|---|
| Applying concepts | 10 |
| Communication | 3 |

|!|BCEEC0110000003|!|

Explain how plants absorb water from roots and then transport it to the leaf.

(11 marks)

##

| Content | Description | Mark |
|---|---|---|
| Water absorption | - Soil water usually has a water potential higher than that of the cytoplasm and the cell sap of the root hair cells.<br>- Water therefore moves into the root hair by osmosis.<br>- Water moves across the cortex from cell to cell by osmosis until it reaches the xylem vessels.<br>- Some water may move inwards along the cell walls, from cell to cell, without entering the cytoplasm or the vacuole.<br>- Eventually, water is drawn up the xylem vessels by the transpiration pull. | (Max. 4) |
| Water transport | - In the leaves, water evaporates from the surface of mesophyll cells through the stomata by transpiration.<br>- Water is continuously drawn from neighbouring cells to replace the water lost.<br>- Eventually, water is drawn from the xylem vessels.<br>- A water potential gradient is generated, causing water to flow from the xylem through leaf cells to replace the water loss.<br>- Due to the surface tension of water, a force (transpiration pull) is produced which pulls water up the xylem vessels from the root. | (Max. 4) |
| **Communication** | | (Max. 3) |

(11 marks)

##

| Skill assessed | Marks |
|---|---|
| Understanding basic concepts | 8 |
| Communication | 3 |

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